A Certified Negative Finite-Window Weil Witness for the Davenport–Heilbronn Function, with a Numerically Positive Remaining-Zero Form

Elias De Jesús
Independent Researcher
ORCID 0009-0007-0190-9143
dejesuselias10@gmail.com

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Abstract

The Davenport–Heilbronn function f satisfies a ζ-shaped functional equation, is entire after completion, has all its zeros in an explicit strip, and has no Euler product; it is known to have zeros off the critical line. It is therefore the natural negative control for any finite-window Weil positivity detector. We exhibit one explicitly specified real test function g_* supported in $[-2, 2]$, odd, in a 37-dimensional span, whose Davenport–Heilbronn Weil value is enclosed in ball arithmetic as $Q_f(g_*)/\|g_*\|_2^2 = -0.679071346441480590986377586093\ldots$ with radius 1.39×10^{-116}, and a second such function at $L = 1.8$ with value $-1.68123895523670103\times10^{-16}$. Both are strictly negative. Decomposing against a census of the zeros to height 200, the single off-line quartet at $\beta = 0.808517182456637386$, $\gamma = 85.6993484853775922$ contributes -0.985198326439305, and the entire remaining spectrum contributes $+0.306126979997825$; and the minimum eigenvalue of the remaining-zero form is positive on every finite space searched, at every scanned window and sector, collapsing to 8.47×10^{-36} at $L = 2.6$ while the full form is at -2.14. That positivity is *numerical and finite-dimensional*: it is a statement about Galerkin restrictions, not about the whole-window form, and the decomposition reads the census zeros as actual zeros, which would need certified root localization that is not supplied here. The computations therefore provide numerical evidence for positivity of the remaining form on the tested finite spaces; the hypothesis needed for a whole-window application of the remaining-zero theorem of the companion paper remains unproved. On the evidence available the negativity is attributable to the off-line quartet O_1. What this rules out is any claim that finite-window Weil positivity follows from a functional equation, entireness and a zero-containing strip; it says nothing about ζ, and it cannot separate "has off-line zeros" from "has no Euler product".

0. Non-claims

Stated first, not in an appendix.

1. **Nothing about RH or GRH.** Every object here is the Davenport–Heilbronn function. It is not ζ, it is not an L-function of anything, and no statement below transfers to either.
2. **f has no Euler product, and nothing here transfers to ζ.** A detector that sees this function's off-line zeros is a detector that is not blind. It is not thereby a detector that would see ζ's, if ζ had any.
3. **The confound, stated here and not buried.** f differs from ζ in *two* ways at once: it has zeros off the critical line, and it has no Euler product. Every measurement below is

consistent with either being the cause of the negativity, and nothing here separates them. Separating them would need an object with one property and not the other, and no such object is known to the author. In particular, the observation of §5 that the remaining-zero form is positive is evidence *internal to this object* that the off-line zeros carry the negativity; it is not a proof, because the remaining-zero form is itself built from a function with no Euler product.

4. **Tiers.** Every zero location is [T2]. The two witnesses of §4 are [T2-C]. The onset scan of §6 is [T2-N]. **Non-detections certify nothing**: a finite basis that finds no negative direction is not evidence of positivity, and the absence of a witness at $L = 1.6$ is reported as an absence, never as positivity.

5. **The $L = 1.8$ onset is not certified.** A negative witness *at* $L = 1.8$ is certified (§4); that the form is positive at $L = 1.6$ is not, so the crossing is bracketed by one certified negative and one non-detection, and the location of the crossing remains [T2-N].

6. **The two literature witnesses are not replayed.** The first has since been read in full and is cited as [8]; for the second only a landing page was ever seen. Neither set of numerical certificates was re-executed here. See Appendix C. No quantity in this note was fitted to either.

7. **A finite positive matrix proves positivity only on its own test space.** The positivity of the remaining-zero form in §5 is a statement about a 37-dimensional restriction at each window, never about the whole finite-window function space.

8. **Numerical root residuals do not establish census completeness**, and agreement with a published decimal is a cross-check, not an interval certificate.

9. **Not treated here.** The degenerate case $\gamma = 0$ of the orbit contribution, where the parity split behaves differently, is the subject of a companion note and is not used anywhere below.

Epistemic tiers

Tier	Meaning	Discharged by
T1	exact or proved	a proof, or exact arithmetic over $\mathbb{Z}$ or $\mathbb{Q}$
T2	numerical	stated working precision + a convergence or noise-floor check
T2-C	certified arithmetic	an Arb ball enclosure with a reproducible certificate
T3	interpretation	argument from T1/T2 evidence, labelled as such

T2-C is *not* a formal-proof assistant verification and is nowhere described as one. Every number printed in this note that is *ours* is re-derived from the data files by `note/verify_note_numbers.py`, which also fails if the note prints more digits than its source carries. Numbers attributed to other authors are a separate category: they are not re-derived here, and the verifier instead requires each to appear verbatim in `SOURCE_AUDIT.md`, so that it is traceable to a reading of the source rather than typed from memory. Nothing in this note replays another author's certificate.

1 The function

Let χ be the character modulo 5 with $\chi(0), \dots, \chi(4) = 0, 1, i, -i, -1$, primitive and odd. Put

$$f(s) \;=\; \tfrac{1}{2}\big[(1 - i\kappa)L(s, \chi) + (1 + i\kappa)L(s, \bar{\chi})\big]. \tag{1}$$

Davenport and Heilbronn [4] introduced this function and showed it has zeros off the critical line; Titchmarsh [9, §10.25] is the standard exposition.

κ from the Gauss sum [T1]. Write $\varepsilon_\chi = \tau(\chi)/(i\sqrt{5})$, so that $\Lambda(s,\chi) = \varepsilon_\chi \Lambda(1-s,\bar\chi)$ with $\Lambda(s,\chi) = (5/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})L(s,\chi)$. Substituting (1) into $\Lambda_f(s) = (5/\pi)^{(s+1)/2}\Gamma(\frac{s+1}{2})f(s)$ and demanding $\Lambda_f(s) = \Lambda_f(1-s)$, the coefficient of $\Lambda(s,\chi)$ matches precisely when

$$(1 - i\kappa)\,\varepsilon_\chi \;=\; 1 + i\kappa, \qquad \text{i.e.} \qquad \kappa = \tan\!\big(\tfrac{1}{2}\arg\varepsilon_\chi\big), \tag{2}$$

and the coefficient of $\Lambda(s,\bar\chi)$ then matches automatically because $\varepsilon_{\bar\chi} = \overline{\varepsilon_\chi}$. Note that (1) is self-reciprocal while the equation for $L(s,\chi)$ alone exchanges χ and $\bar\chi$; that exchange is exactly what (1) is built to undo. Evaluating, $\tan\arg\varepsilon_\chi = \sin 36^\circ/\sin 72^\circ = 1/\varphi$ with φ the golden ratio, whence the closed form

$$\kappa \;=\; \frac{\sqrt{10-2\sqrt{5}}-2}{\sqrt{5}-1} \;=\; 0.28407904384041229603\ldots$$

Our κ is computed as $\tan(\frac{1}{2}\arg\varepsilon_\chi)$ from the Gauss sum, not imported; it agrees with the closed form to 3.9×10^{-62}, and the residual of (2) is exactly 0 at the working precision. The Dirichlet coefficients are $(a_1,\ldots,a_5) = (1,\kappa,-\kappa,-1,0)$, real and periodic but *not* multiplicative: $a_2a_3 = -\kappa^2$ while $a_6 = a_1 = 1$.

Functional equation, measured [T2]. At twenty reproducibly chosen points avoiding the critical line, the real axis and the gamma poles, the scale-aware relative residual of $\Lambda_f(s) = \Lambda_f(1-s)$ is at worst 6.951×10^{-58} at 60 working digits and 8.092×10^{-99} at 100; it tracks the precision, which is the check that it is a precision floor and not a defect. The relative form is necessary: one of the twenty points has $|\Lambda_f| \approx 2\times10^{-31}$, where an absolute residual would be meaningless.

Strip [T1]. For $\sigma \ge 2$, $|f(s)-1| \le \sum_{n\ge2}|a_n|n^{-\sigma} \le \zeta(2)-1 = 0.644934066848\ldots < 1$. Sharpening the same elementary estimate with the actual $|a_n|$ gives the abscissa σ_0 with $\sum_{n\ge2}|a_n|n^{-\sigma_0} = 1$, namely $\sigma_0 = 1.395136158235$, so every zero of Λ_f lies in $1-\sigma_0 < \operatorname{Re} s < \sigma_0$. The strip is wide — this function has zeros with $\operatorname{Re} s > 1$ — and it is what makes the census contour of §3 meaningful without an arbitrary cutoff.

Coefficients of $-f'/f$ [T1] and [T2]. In $\operatorname{Re} s > 2$ set $-f'(s)/f(s) = \sum_{n\ge2} b_n n^{-s}$, so that $b_n = a_n\log n - \sum_{d\mid n,\,2\le d<n} b_d a_{n/d}$ with $b_1 = 0$. Exactly, $b_2 = \kappa\log 2$, $b_3 = -\kappa\log 3$, and

$$b_6 \;=\; (1+\kappa^2)\log 6 \;=\; 1.9363560766210387863,$$

nonzero at a non-prime-power. For ζ the set of $n < 50$ that are not prime powers and have $b_n \neq 0$ is empty; here it has 15 members. The missing Euler product is therefore visible in the coefficients the detector consumes, not only in the zeros. A second computation of b_n through the Dirichlet inverse of (a_n) agrees with the recursion to 1.245×10^{-60} for $n \le 200$.

2 The window form

Let g be real and supported in $[-L,L]$, and set

$$G(z) = \int g(u)e^{zu}\,du, \qquad h(v) = \int g(u+v)g(u)\,du, \qquad H(z) = G(z)G(-z),$$

so h is supported in $[-2L,2L]$ and $H(it) = |G(it)|^2$. The spectral form runs over the zeros of the *entire completed* function, $Q_f(g) = \sum_\rho \operatorname{mult}(\rho)H(\rho-\frac{1}{2})$, and all reported values are normalised

as $Q_f(g)/\|g\|_2^2$. Moving the contour as in the companion paper's §4–5, with the conductor 5 and the odd archimedean factor $\psi(\frac{3}{4}+\frac{it}{2})$ in place of the ζ data, gives

$$Q_f(g) \;=\; h(0)\log\frac{5}{\pi} \;+\; \frac{1}{2\pi}\int_{\mathbb{R}} |G(it)|^2\,\mathrm{Re}\,\psi\Big(\tfrac{3}{4}+\tfrac{it}{2}\Big)\,dt \;-\; 2\sum_{2\le n\le e^{2L}} \frac{b_n\,h(\log n)}{\sqrt{n}}. \tag{3}$$

There is **no pole term**: Λ_f is entire, the trivial zeros of f at $s=-1,-3,\dots$ cancelling the poles of the gamma factor. The arithmetic sum is **finite** because h vanishes beyond $2L$; at $L=2.6$ the cutoff is $n\le 181$ with 122 nonzero terms, and the bound is decided in integer arithmetic and then re-tested as $\log n < 2L$ at full precision, never by a binary float. The formula is of the Weil–Yoshida–Bombieri type [10, 2]; what is specific to f is that b_n is not supported on prime powers.

Zero contributions. An on-line conjugate pair $\{\frac{1}{2}\pm i\gamma\}$ contributes $2H(i\gamma)=2|G(i\gamma)|^2\ge 0$. The complete orbit of a non-real off-line zero, $O=\{\beta\pm i\gamma,\,1-\beta\pm i\gamma\}$, contributes, with $\delta=\beta-\frac{1}{2}$,

$$Q_O(g) \;=\; 4\,\mathrm{Re}(A\overline{B}) \;=\; 4(|P|^2-|S|^2), \qquad A=G(\delta+i\gamma),\; B=G(-\delta+i\gamma), \tag{4}$$

where $P=\int g\cosh(\delta u)e^{i\gamma u}du$ and $S=\int g\sinh(\delta u)e^{i\gamma u}du$. This is a *signed* rank-two form, with a positive cosh channel and a negative sinh channel, and it is not $|G(\delta+i\gamma)|^2$. As $\delta\to 0$ it tends to $4H(i\gamma)$, the contribution of *two double* on-line zeros, so multiplicity is preserved: four zeros in, four out.

Gate A [T2-N]. The arithmetic side (3) is compared with two independent evaluations and three exact-algebra controls. The contour route writes $Q_f(g)=(2/\pi)\,\mathrm{Re}\int_0^\infty(\Lambda_f'/\Lambda_f)(\sigma_c+it)H(\sigma_c-\frac{1}{2}+it)\,dt$ at $\sigma_c=1.42$, to the right of every zero, with Λ_f'/Λ_f built from f and f' directly and *never* from the b_n; the zero route sums over the census of §3.

Check	Discrepancy	Declared budget	Tier
arithmetic vs. logarithmic-derivative contour	1.73×10^{-13}	1.316×10^{-6}	T2-N
arithmetic vs. zero-side sum over the census	5.65×10^{-11}	7.13×10^{-5}	T2-N
control: negate the selected quartet (sign flip)	1.83×10^{-62}	exact algebra	T1
control: omit b_6 at a window containing $n=6$	1.94×10^{-61}	exact algebra	T1
control: flip each arithmetic term in turn	3.11×10^{-61}	exact algebra	T1

The quartet control negates the selected orbit's contribution on the zero side alone and checks that the discrepancy moves by exactly twice it; with $Q_{O_1}(g)=8.720121523693614\times10^{-11}$ on that low-frequency test function, the predicted and observed changes agree to 1.83×10^{-62}. The b_6 control predicts a change of $+2b_6h(\log 6)/\sqrt{6}=-0.379255712825982$ from $b_6=1.93635607662104$ and $h(\log 6)=-0.239879170384883$, and observes it to 1.94×10^{-61}. The term-level sign flip negates each of the archimedean, conductor and arithmetic terms in turn; the conductor row is the control that would catch a wrong conductor, and it reproduces $-2\log 5=-3.21887582486820$ for a unit-normalised test function in every sector and window. **Classification: PASS-N** — converged numerical agreement, explicitly uncertified, because the quadrature in the contour route and the zero density in the zero-side tail are a library estimate and a measured quantity, not rigorous bounds. The three controls, which are what would catch a wrong factor, a wrong conductor or a dropped term, are exact.

3 The zero census to $T = 200$

Counting by the argument principle on the strip $-0.4 < \operatorname{Re} s < 1.4$, whose vertical edges are provably zero-free by §1, gives 130 zeros with $0 < \operatorname{Im} s < 200$ against a Riemann–von Mangoldt main term of 129.548; the worst deviation of a winding number from an integer over the whole ladder is 1.32×10^{-23}. Sign changes of the Hardy-type Z give 122 critical-line ordinates, and four off-line zeros with $\operatorname{Re} s > \frac{1}{2}$ are located by box subdivision and Newton, each carrying local winding number exactly 1 on a circle of radius 0.02, hence simple. The reconciliation

$$122 \;+\; 2 \cdot 4 \;=\; 130$$

is exact, and it is also what closes the band $\frac{1}{2} < \operatorname{Re} s < 0.55$ that the located search deliberately avoids: an unlocated zero there would break it.

	β	γ	$\delta = \beta - \frac{1}{2}$	$\lvert f(\rho)\rvert$	winding
O_1	0.808517182456637386	85.6993484853775922	0.308517182456637386	1.75×10^{-60}	1
O_2	0.650830080609737082	114.163342730756981	0.150830080609737082	3.21×10^{-60}	1
O_3	0.574356050450805991	166.479305913168156	0.0743560504508059907	5.99×10^{-60}	1
O_4	0.724257694626809780	176.702461242855825	0.224257694626809780	2.99×10^{-60}	1

The residuals are at the working precision of the census run (60 digits); at 140 digits the refined O_1 has $|f(\rho_1)| = 8.2 \times 10^{-141}$ with Newton step 6.5×10^{-141} and is stable to 3.17×10^{-106} between 100 and 140 digits. These are numerical approximations to the roots, refined until the residual $|f(\rho)|$ and the Newton step sit at the working precision floor and the location is stable under raising that precision. They are not interval certificates: no enclosure is computed, so the identification of these points as zeros of f rests on those residuals rather than on certified root localization.

Which are cross-checked [T2]. One of the four, and only against a secondary source. O_1 was found from a decimal seed supplied with the task; Newton moved that seed by 7.8×10^{-15}, so the seed's digits are correct to about fourteen places. The provenance of that decimal is a secondary report of the literature computations of Spira [7] and Balanzario and Sánchez-Ortiz [1]; **neither paper was read** — the AMS PDF of the latter returned HTTP 403 from the machine used — and no numerical value from either appears in this note. O_2, O_3 and O_4 are **not** cross-checked against anything: they are this computation's own. Bombieri and Ghosh [3] is the standard modern reference for the analytic theory of f; it was not read either, and no result or constant of theirs is used — in particular the counting function above is the computed argument-principle count, with the main term quoted only as a diagnostic.

4 The certified witnesses

[T2-C] Two test functions are frozen, with every coefficient stored as an exact 30-digit decimal string. Both are odd, both are in the span of a low block together with a block resonant at γ_1, and the Gram matrix of the normalised basis is the identity exactly (deviation 4.83×10^{-60} and 7.4×10^{-60}), so the generalised eigenproblem is the ordinary one and no direction was regularised or discarded. The coefficients of g_* are in Appendix A.

	g_*	$g_\sharp$
L	2.0	1.8
sector	odd	odd
N	37	57
active arithmetic range	$n \le 54$, 39 nonzero terms	$n \le 36$, 26 nonzero terms
enumeration cap	$n \le 56$	$n \le 38$
$Q_f/\|g\|_2^2$	-0.679071346441480590986	$-1.68123895523670103 \times 10^{-16}$
ball radius	1.39×10^{-116}	8.15×10^{-177}
strictly negative	yes	yes

The two rows are not the same thing. A term can be nonzero only while $\log n < 2L$, so the *active* range is $n \le \lfloor e^{2L} \rfloor$; the certificate enumerates two indices beyond it as a margin, and those contribute nothing. Quoting the cap as if it were the active cutoff would overstate the arithmetic actually entering the form.

The $L = 1.8$ witness reproduces an independent result. [T3] Ziad [8] certifies a negative Davenport–Heilbronn value at $a = 9/5$ on odd test functions, in a convention whose a is this note's L (Appendix C). His witness is constrained by $u \perp \sinh(x/2)$ and lives in a 140-mode sine basis; $g_\sharp$ is unconstrained and lives in the 57-dimensional span above. Both are odd, both are certified, and both are negative at the same window, so $g_\sharp$ is an *independent reproduction of negativity at* $L = 1.8$. The two Rayleigh values are normalized evaluations of the *same* Davenport–Heilbronn form at the *same* window, so they may be compared directly *as witness values*: both are strictly negative, and his is the deeper, hence the stronger bound. Each value bounds the infimum from above over *every* admissible space containing its witness, the full window included: $\lambda_{\text{full}} \le \lambda_{\text{trial}} \le Q(u)/\|u\|_2^2$. What they are not is equivalent *approximations* to the window minimum: they are evaluations on different finite trial spaces, neither of which is known to contain a minimizer, so neither value says how close the true infimum lies.

Why the certificate is complete. The enclosure is of the *whole* quantity, not of an approximation to it, and there is nothing outside the ball:

- **The arithmetic sum is finite and complete.** h vanishes beyond $2L$, so (3) has 39 terms at $L = 2$ and no tail is dropped.
- **No quadrature.** The archimedean term is the closed form in digamma, the Hurwitz zeta and the Lerch transcendent; Arb encloses all three.
- **No truncation and no spectral input.** $Q_f(g_*)$ is computed from the arithmetic side alone. It does not know where any zero is, so the census plays no part in it and the sign does not depend on the census being right.
- **The test function carries no error.** Its coefficients are exact finite decimals; the ball encloses Q_f for *this* g_*, not for a rounding of some other one.
- **Independent implementation.** The certifying code re-implements the cross-correlations, κ, the b_n and the archimedean closed form against Arb, sharing no code with the `mpmath` assembly; the two agree to all 44 digits `mpmath` carries, so the certificate is simultaneously a cross-check of both.

The only inputs are Arb's enclosures and the derivation of (3). κ enters through square roots only. This is [T2-C] and not a formal verification.

5 The decomposition

[T2-C] Both terms below are finite sums and both are enclosed. At g_*:

$$\underbrace{-0.679071346441480590986}_{Q_f(g_*)} = \underbrace{-0.985198326439305481794}_{Q_{O_1}(g_*)} + \underbrace{+0.306126979997824890807}_{Q_{\mathrm{rest}}(g_*)}, \tag{5}$$

with ball radii 1.39×10^{-116}, 4.06×10^{-117} and 1.63×10^{-116}, and additive reconciliation residual 2.99×10^{-116}. Here $Q_{\mathrm{rest}} = Q_f - Q_{O_1}$ is evaluated by subtracting the exact rank-two orbit form (4) from the arithmetic side, so *every* remaining zero — including every off-line zero above the census height and every zero never located — is included implicitly. Nothing is replaced by a finite zero list.

The selected quartet therefore carries more than the whole of the negativity, and the rest of the spectrum returns 31.07% of it. At $g_\sharp$ the same decomposition reads $-6.45917960933013 \times 10^{-14} + 6.44236721977777 \times 10^{-14}$: near the onset the cancellation is 99.7397%, which is why the witness there is of order 10^{-16} rather than 10^{-14}.

The remaining-zero form is positive on every space searched [T2-N]. $\lambda_{\min}$ of the restriction of Q_{rest} to the level-2 space, alongside $\lambda_{\min}(Q_f)$:

L	$\lambda_{\min}(Q_f)$, even	$\lambda_{\min}(Q_{\mathrm{rest}})$, even	$\lambda_{\min}(Q_f)$, odd	$\lambda_{\min}(Q_{\mathrm{rest}})$, odd
1.0	$4.034840454024 \times 10^{-5}$	$4.03625203581 \times 10^{-5}$	0.01356156739650	0.0132281100537
1.2	$8.780065301754 \times 10^{-9}$	$8.46979844459 \times 10^{-9}$	$9.702616267333 \times 10^{-6}$	$9.54583799747 \times 10^{-6}$
1.4	$5.087177960475 \times 10^{-14}$	$4.64125898422 \times 10^{-14}$	$6.538829304189 \times 10^{-11}$	$6.63381728826 \times 10^{-11}$
1.6	$2.549827775205 \times 10^{-19}$	$2.25086996550 \times 10^{-19}$	$2.153851664770 \times 10^{-16}$	$2.21332806419 \times 10^{-16}$
1.8	$7.580609464352 \times 10^{-24}$	$7.01854463036 \times 10^{-24}$	$1.007541951495 \times 10^{-20}$	$1.08891994498 \times 10^{-20}$
2.0	-0.4896520626161	$2.02946591361 \times 10^{-27}$	-0.6790713464415	$1.20623694960 \times 10^{-24}$
2.2	-1.219875374653	$1.22467880540 \times 10^{-30}$	-1.178435305399	$4.87721021342 \times 10^{-28}$
2.4	-1.503014486764	$1.81134931584 \times 10^{-33}$	-1.590268954821	$4.26137491134 \times 10^{-31}$
2.6	-2.137655567743	$8.47226605297 \times 10^{-36}$	-2.005507257484	$8.13613127375 \times 10^{-34}$

All 18 rows are positive. Once Q_f is macroscopically negative the remaining-zero form does not merely stay positive: it collapses towards zero, reaching 8.47×10^{-36} at $L = 2.6$ while Q_f is at -2.14. Removing O_1 from the frozen witness's space and re-optimising gives $\lambda_{\min} = +1.20623694960 \times 10^{-24}$; no other negative direction is found there, which is a statement about that space and not about positivity.

The consequence, stated exactly. The remaining-zero theorem of the companion paper assumes a remaining-zero form that is nonnegative, and states that assumption as *sufficient*, not necessary, for its conclusion. On this object the computations give **numerical evidence** for that hypothesis on every finite space searched. That is not the hypothesis itself, and it does **not** entitle one to apply the theorem here: a whole-window application needs $Q_{\mathrm{rest}} \geq 0$ on the whole window, which remains unproved. What can be said is that the theorem's proof uses the hypothesis through that single inequality alone, so nothing further about where the other zeros lie would be needed were it available. The masking contribution is 31.07% in the witness direction and does *not* reverse the sign. What is *not* established is that a positive finite restriction establishes the whole-window hypothesis: controlling the infinite-dimensional complement and its coupling to the block would be needed, and a positive Galerkin matrix supplies neither.

What the quartet term is, exactly. [T3] The decomposition evaluates the Weil pairing at the census ordinates. Reading the resulting number as *the contribution of an actual zero* of f requires knowing that a zero of f really sits at that (β, γ), to the precision used — a certified root localization. The census of §3 locates its ordinates by adaptive sampling with winding numbers integral to 1.32×10^{-23}, which is strong numerical evidence and not an interval certificate. So the quartet term is certified *as an evaluation of the pairing at a stated point*, and its identification with a zero of f inherits the census's status, not the witness's.

What Q_{rest} contains. Deleting the quartet O_1 leaves *every other zero* of f, and that includes the remaining off-line quartets of the census — O_3 among them — not only the on-line ones. Q_{rest} is therefore not a "critical-line form"; it is the full form with one quartet removed.

And the confound, once more. Q_{rest} is built from the zeros of an object with no Euler product, and it is positive on the finite spaces searched; that is the strongest evidence *internal to this object* that the negativity is attributable to O_1, and it is not a proof.

6 Onset

[T2-N] The scan runs $L = 1.0, 1.2, \ldots, 2.6$ in both parity sectors at three declared basis levels; enrichment can only lower $\lambda_{\min}$, and the levels are reported as enrichments, never folded into one another. Over all 54 rows the Rayleigh quotient of the returned eigenvector agrees with the eigenvalue to 5.76×10^{-60}, so the eigensolver is confirmed by direct evaluation. At the deepest level, $\lambda_{\min}$ falls by a factor 1.459×10^{17} from $L = 1.0$ to $L = 1.6$ while remaining positive, and then changes sign:

L	even, lvl 1	even, lvl 2	even, lvl 3	odd, lvl 1	odd, lvl 2	odd, lvl 3
1.6	$1.97261814306 \times 10^{-13}$	$2.54982777520 \times 10^{-19}$	$2.66141166363 \times 10^{-22}$	$2.91871108168 \times 10^{-11}$	$2.15385166477 \times 10^{-16}$	$1.09593824675 \times 10^{-18}$
1.8	$2.91410423267 \times 10^{-15}$	$7.58060946435 \times 10^{-24}$	$-2.66180729007 \times 10^{-21}$	$1.19062366907 \times 10^{-13}$	$1.00754195149 \times 10^{-20}$	$-1.68123895524 \times 10^{-16}$
2.0	-0.188728484539	-0.489652062616	-0.547390086184	-0.658652328928	-0.679071346441	-0.710175512230

Twenty-six of the 54 rows are negative. The crossing is not gradual: the form sits on a knife edge through most of the scanned range and then falls off it, from -1.68×10^{-16} at $L = 1.8$ to -0.71 at $L = 2.0$.

Which sector detects first. The scan does not resolve this. Both sectors are positive at $L = 1.6$ and negative at $L = 1.8$ at level 3, so on this grid they cross together; the odd value is deeper by a factor 6.316×10^4 at $L = 1.8$ and by 0.16 at $L = 2.0$. Since $K_{\text{even}}(2.0, \gamma_1)/K_{\text{odd}}(2.0, \gamma_1) = 1.00634$ at this height, the two sectors have essentially the same isolated sensitivity, and **no structural explanation is offered** for the depth difference.

Two comparisons. Both are computed by the verifier from the Phase-A values of (δ_1, γ_1), not quoted from elsewhere.

(i) Bandwidth scale. With $L_{\text{pred}} = \frac{1}{2}\log(5\gamma_1/2\pi) = 2.11120303$, and the measured onset bounded above by the certified negative witness at $L = 1.8$ — a negative value at a window is an *upper* bound for the first loss of positivity, and the macroscopic value at $L = 2.0$ is a second, weaker upper bound, not a lower one —

$$\frac{L_{\text{pred}}}{1.8} = 1.17289, \qquad \frac{L_{\text{pred}}}{2.0} = 1.0556.$$

This is a reference scale, not a predicted onset: its provenance is a bandwidth heuristic that was not verified, and the bracket is one-sided in the sense of non-claim 5.

(ii) The δ^2 coefficient. In the odd sector the exact extremal constant is $K_{\mathrm{odd}}(L,\gamma) = \int_{-L}^{L} u^2\cos^2(\gamma u)\,du = 2.6582355735990895$ at $L = 2$, $\gamma = \gamma_1$, and the companion paper's δ^2 coefficient gives

$$4K_{\mathrm{odd}}(2.0,\gamma_1)\,\delta_1^2 \;=\; 1.01207377, \qquad \text{against} \qquad |Q_{O_1}(g_*)| = 0.985198326439305,$$

a ratio of 1.02728. Measured instead against the actual drop from the collapsed configuration, $Q_{O_1}(0) - Q_{O_1}(\delta_1) = 0.986028353$ with $Q_{O_1}(0) = 4|\hat{f}(\gamma_1)|^2 = 0.000830026353$, the ratio is 1.02641.

Is that inside what the remainder allows? Yes, and *so comfortably that the comparison is not a sharp test of the bound.* At $L\delta_1 = 0.617034$ the companion paper's δ^4 allowance is $(\frac{2}{5} + \frac{16}{3}e^{2L\delta})L^5\delta^4 = 5.42744$, while the observed discrepancy $|4K\delta^2 - \text{drop}|$ is 0.02605 — smaller by a factor 208.4. The 2.6% agreement is therefore *better* than the bound requires, by more than two orders of magnitude, and should not be read as confirming the bound's constant; the bound is simply very loose at this displacement. $L\delta_1 = 0.617$ is far outside any small-displacement regime, and no small-δ law is used anywhere in §4 or §5.

7 What this rules out, and what it does not

Ruled out. Any claim that finite-window Weil positivity follows from a functional equation of ζ's shape, together with entireness of the completion and a zero-containing strip. The Davenport–Heilbronn function has all three — (1) is self-reciprocal, Λ_f is entire, and every zero lies in $|\operatorname{Re} s - \frac{1}{2}| < 0.895136$ — and its window form at $L = 2.0$ is negative, at -0.679071346441, now by a *certified example at a stated window* rather than by citation of the existence of off-line zeros.

What this example does *not* rule out. An earlier version of this paragraph offered the same computation as a refutation of the claim that a positive finite Galerkin restriction of Q_{rest} establishes a whole-window hypothesis, on the ground that $\lambda_{\min}(Q_{\mathrm{rest}}) > 0$ on every space searched while Q_f is macroscopically negative. That inference is invalid and is **withdrawn**: since $Q_f = Q_{\mathrm{rest}} + Q_{O_1}$, the form Q_f can be negative while Q_{rest} is positive on the whole window — a configuration compatible with the computations of §5, whose whole-window positivity remains unresolved. The negativity of Q_f therefore carries no information about whole-window positivity of Q_{rest}. The general limitation stands on its own and needs no example: a positive finite restriction does not establish positivity on the whole space, because the infinite-dimensional complement and its coupling to the block are left uncontrolled.

Already known on the positive side. The negativity reported here sits above a *proved* positive range for the same object. Ziad [8] establishes $q_a^{\mathrm{DH}}[u] > 0.4289\,\|u\|_2^2$ for $0 < a \le \frac{1}{2}\log 2$, and his derivation (his §6.2) uses only that u is odd — oddness forces $\int u = 0$, which annihilates the integrated autocorrelation, and the orthogonality constraint he otherwise imposes is not used at that step. It therefore applies to the *full* odd sector studied here, not only to his constrained subspace. Together with the present witnesses this gives a broad bracket for Davenport–Heilbronn on odd test functions: positive for $L \le \frac{1}{2}\log 2$, negative at $L = 1.8$ and $L = 2.0$. It leaves untouched the question this note's scan actually runs into, which is positivity near $L = 1.6$; that window lies far above $\frac{1}{2}\log 2$ and is a separate problem.

Not ruled out, and not claimed. Nothing about ζ: no statement here has a ζ analogue, and the whole point of the control is that it runs where positivity is known to fail. Nothing that separates the two properties of f, off-line zeros and the absent Euler product, which is the confound of non-claim 3. Nothing about other L-functions, other conductors, other bases, or other windows than the nine scanned. Positivity at the scanned $L \le 1.6$ is *not* settled by

this note: those rows are non-detections, not certificates. A broad two-sided bracket for the onset does now exist, but it comes from the positive base above, not from here: positivity for $L \le \frac{1}{2}\log 2$ together with the negative witness at $L = 1.8$ confines the first loss of positivity to $(\frac{1}{2}\log 2,\ 1.8]$. What remains unresolved is positivity *at* $L = 1.6$, and more generally the location of the onset inside that interval; pinning it would need a certified lower bound over a whole space at some window above $\frac{1}{2}\log 2$, which is a different and harder object than a certified value at a point. The two witnesses here are both upper bounds for the onset.

Open, in four sentences. Whether $\lambda_{\min}(Q_{\mathrm{rest}})$, already at 8.47×10^{-36} at $L = 2.6$, stays positive as the basis grows, or whether its near-null direction is the leading edge of a genuine negative one. Whether the broad bracket $(\frac{1}{2}\log 2,\ 1.8]$ of §7 can be narrowed — most directly by proving positivity at $L = 1.6$, which would move its lower end there and leave the onset inside a single grid step. Whether the census can be made rigorously complete, the winding numbers being integers to 1.32×10^{-23} by adaptive sampling rather than by interval arithmetic. And whether an off-line quartet of small δ at larger height — O_3 has $\delta = 0.0743560504508059907$ at $\gamma = 166.479305913168156$ — is seen earlier or later than O_1 by a basis resonant at *its* height, which the present scan, resonant only at γ_1, cannot say.

A The frozen witness g_*

$L = 2.0$, odd sector, $N = 37$. The test function is $g_*(u) = \sum_m c_m\, \phi_{k_m}(u)$ with $\phi_k(u) = \sin((k+1)\pi u/L)/\sqrt{L}$ on $[-L, L]$ and zero outside, so that $\|g_*\|_2^2 = \sum_m c_m^2$. The index set is the low block $k = 0, \dots, 15$ together with the resonant block $k = 44, \dots, 64$, chosen so that $w_k = (k+1)\pi/L$ brackets γ_1. The coefficients are exact finite decimals, read back from the certificate as strings; the table below is generated mechanically from `certificates/witness.json` and is regenerated and compared by the verifier.

m	k_m	c_m	m	k_m	c_m
0	0	−0.00260883678800775017628484317 74	19	47	0.01607940834998211719532609419 76
1	1	0.00593672979836836786761437123 06	20	48	0.00498986780562686767195561151428
2	2	−0.01855340627802310239212618961 05	21	49	0.00002770539295216541778148825863 19
3	3	0.00544846305151052127579958739366	22	50	−0.02221792672610145516464367644 75
4	4	−0.01563960173739446356696957109136	23	51	0.03958684583427772676410956413 4
5	5	−0.01097831646551123906557891880 25	24	52	0.09423029991062382558587280479 73
6	6	−0.01259077543132861586266017474 24	25	53	0.80418741194959654561729049492 9
7	7	−0.01446280489492370934767268368 75	26	54	−0.55999299280330206113217385671 2
8	8	−0.01463439148944907988812762289 96	27	55	0.13770885858325854465943856656 1
9	9	0.01038985243912394370989743498 74	28	56	0.02217767266746974247666347857 07
10	10	0.00448607789902456302212489057999	29	57	0.01504549554032073380837946674 82
11	11	0.02419738638304006093097472390 95	30	58	−0.03030980244999840959813376801 34
12	12	−0.01415455807125121991946130198503	31	59	0.02739276170837229818633895678 13
13	13	0.00838291335579069185266302436363	32	60	0.00112786440916397755739920517311
14	14	0.00306472453668078854500166029552	33	61	−0.02262898011218972818071117886 14
15	15	0.03197189892230664045504957 318	34	62	0.00211860269767553369614679144869
16	44	−0.03118933049139355542929 87951 3548	35	63	0.01958944126354641170481363260 5
17	45	−0.00554065734427570197610267400465	36	64	0.03011276058195138841417236230 9
18	46	−0.03041911016213350699432422665 61			

The mass is concentrated on $k = 53$ and $k = 54$, whose frequencies bracket $\gamma_1 = 85.699348485377592$; the witness has normalised overlap 0.9513529401 with $u\cos(\gamma_1 u)$, the extremiser of the isolated derivative-evaluation functional, while the ground state of Q_{rest} is orthogonal to it to 9.07×10^{-12}.

B Retraction log

Six corrections were made during this work. They are reproduced from `RETRACTIONS.md`, in order.

R1. Two-sided zero-count normalisation. The Riemann–von Mangoldt main term was written $(T/\pi)\log(5T/2\pi e)$, which counts $|\operatorname{Im}\rho| < T$, while the census counts $0 < \operatorname{Im}\rho < T$. Corrected to $(T/2\pi)\log(5T/2\pi e)$. The near-exact factor of two between the winding numbers and the "main term" is what exposed it. Diagnostic only: the counts come from the argument principle, and no reported number changed.

R2. Off-line search contour too close to the critical line. The first implementation subdivided from $\operatorname{Re} s \geq 0.505$, which passes within 0.005 of the on-line zeros; $\arg f$ then swings by nearly π between samples and the adaptive refinement recursed to its depth limit on every band. Corrected by searching at $\operatorname{Re} s \geq 0.55$ and closing the uncovered band by the reconciliation of §3, which is a stronger check and cheap. Method only; no reported number changed.

R3. Division by zero at a removable singularity. The closed form for $\int \cos(wu)e^{zu}du$ has a $0/0$ at $z = \pm iw$. A unit test evaluating there caught it; an explicit limit branch was added. The condition never arose in a reported computation — H is evaluated at zero ordinates, not at basis frequencies — and the branch exists so the test can cover the case.

R4. The first independent archimedean route was too weak to be a check. It evaluated the frequency-side integral by quadrature; the integrand oscillates with period π/L and the result was accurate only to about 10^{-6}, which cannot distinguish a correct assembly from one with a small error. Replaced by Gauss's integral for ψ, giving a one-dimensional non-oscillatory integral with an exact tail. The two routes now agree to 6.225×10^{-21} against an analytic bound of 1.66×10^{-20} on the piece deliberately omitted near the origin.

R5. A gate threshold that could not be met. Phase B required agreement below 10^{-40} while its own declared budget was 10^{-20}. A gate that cannot pass is not a gate; the threshold was tied to the route's own budget. No reported value changed.

R6. Decimal window lengths parsed at the ambient precision. This is the one that mattered. `mpmath`'s `mpf` rounds a decimal string to the *current* working precision and freezes it. A window length built as `mpf('1.8')` before the precision was raised carried only 15 digits, and the subsequent precision ladder at 60, 100 and 160 digits then evaluated the form at that same slightly-wrong L three times, agreed with itself to 10^{-85}, and reported a value wrong in the *thirteenth* significant digit. A ladder that agrees with itself to 85 digits is not evidence that the number is right. It was caught only because two scripts froze the same witness independently and disagreed in the 13th digit — the ladder itself could not have caught it, and would not have, at any precision. Every decimal constant that feeds the form now goes through a parser fixed at 220 digits, and a guard test fails if any of them is reconstructed at the ambient precision. Blast radius: the certified witness sits at $L = 2.0$, exactly representable in binary, and is unaffected; the window scan ran at a consistent precision throughout and is unaffected; the $L = 1.8$ witness was recomputed, and its sign and first twelve digits are unchanged.

C The two literature witnesses

Two contemporaneous documents report negative Davenport–Heilbronn window witnesses. Their status differs, and the difference matters: **the first has now been read in full; for the second only a landing page was ever seen.**

First document — read. Ziad [8]. The PDF was retrieved and read on 2026-11-10 (`SOURCE_AUDIT.md` F1 records the file hash and the located statements). Its framework is odd functions on $J_a = (-a, a)$, zero-extended, with the constraint $u \perp \sinh(x/2)$.

The convention question is settled, and the earlier hypothesis was wrong. $J_a = (-a, a)$ is the support of the *test function*, which is this note's $[-L, L]$. So a corresponds to L directly, with no factor of two, and his $a = 9/5$ is this note's $L = 1.8$. An earlier version of this appendix hypothesised a factor of two between the two documents' window numbers, on the ground that one might be measuring the support of h rather than of g; for Ziad that hypothesis is **withdrawn**. It survives only as an open question about the second document below.

What he establishes, at the scope he states it: a positive base $q_a^{\mathrm{DH}}[u] > 0.4289\,\|u\|_2^2$ for $0 < a \le \frac{1}{2}\log 2$ (his §6.2); a certified negative witness at $a = 9/5$ with $q_{9/5}^{\mathrm{DH}}[u]/\|u\|_2^2 = -6.1689581904177 \times 10^{-9} \pm 4.3 \times 10^{-23}$ (his §6.3); and hence a first loss-of-positivity scale $\frac{1}{2}\log 2 < b_{\mathrm{DH}}^{\natural} < 9/5$. He asserts no numerical construction of the first-kernel vector, and labels his own zero-side anatomy "diagnostic only" (his §6.4). His companion ζ result certifies $0 < \lambda_1 < 5.002295985216 \times 10^{-19}$ at $\frac{1}{2}\log 7$ over the *constrained* odd space — not over every test function at that window, and the bound is that value, not something below 5×10^{-19}.

Relation to this note. His witness is constrained by construction, but a constrained odd function is still an odd function, so it lies in the full odd sector studied here; the finite bases differ (his 140 sine modes against the 57-dimensional span of the $L = 1.8$ witness $g_\sharp$, which is the one at his window). The $L = 1.8$ witness of §4 is therefore best read as an **independent reproduction of negativity at that window** — same object, same window, same convention, different subspace and different test function. Being normalized evaluations of the same form at the same window, the two Rayleigh values may be compared *as witness values*, and each bounds the window infimum from above — over its own trial space and over every larger admissible space containing it. What their differing finite trial spaces forbid is treating them as equivalent *approximations* to the minimum.

Second document — not read. A Zenodo series on cross-energy of Dirichlet partial sums, from which the sentence "positive up to a window $L \approx 3.6 \ldots \lambda_{\min} = -0.73$ at $L = 4$, with a minimizer concentrated on its off-line zero $0.8085 + 85.70i$" was surfaced from a landing page. Nothing below is a reading of its argument. Its window numbers are twice Ziad's, and the factor-of-two hypothesis above — that one document may be measuring the support of h rather than of g — is retained for this document alone. **It is a hypothesis, not a finding**, and until the document is read neither agreement nor disagreement with it can be asserted. Our level-3 value at $L = 2.0$, -0.7101755122300, is in the same region as the quoted -0.73 under the hypothesis, and is an upper bound for the true minimum because it is computed on a finite subspace — but that observation is worth exactly as much as the convention hypothesis it rests on.

Nothing in this note was fitted to either value. The scanned window range was fixed before either document was located, no threshold or basis was tuned, and no novelty is claimed: at least two contemporaneous efforts address the same control, one of them with its own interval certificate.

D Reproduction

Code and data: `https://github.com/innerlightr-wq/weil-window-control`, directory `extensions/davenport-heilbronn-2026-10-23/`. Python 3.14.7 (CPython), `mpmath` 1.3.0, `python-flint` 0.9.0 (Arb). No `numpy`, no `scipy`, and no `float64` on any path reaching a reported value. All conventions, precisions, basis levels, windows and thresholds are in `src/config.py`; no randomness is used.

Result	Script	Data	Runtime
§1 function, κ, functional equation	`src/phaseA.py`	`results/phaseA.json`	≈3 m
§1 b_n, §2 form	`src/phaseB.py`	`data/dh_coefficients.csv`	≈6 m
§2 orbit conventions	`tests/test_orbits.py`	—	≈2 m
§3 census	`src/census.py`	`data/dh_zeros.csv`	≈25 m
§2 Gate A	`src/phaseC.py`, `src/gate_signflip.py`	`data/gate_checks.csv`, `data/gate_signflip.csv`	≈25 m
§6 scan	`src/phaseD.py`	`data/window_scan.csv`	≈45 m
§4 freeze	`src/freeze.py`	`certificates/witness*.json`	≈20 m
§4 certificates	`src/certify.py`	`certificates/ball_certificate*.txt`	≈2 m
§5 remaining form	`src/phaseE.py`	`data/remaining_form.csv`	≈15 m
§5 masking	`src/phaseF.py`	`data/masking.csv`	≈5 m
§6 K, perturbation	`src/phaseG.py`	`data/sensitivity.csv`	≈10 m

The two certificates are reproduced by

```
.venv/bin/python src/certify.py 400
.venv/bin/python src/certify.py 600 certificates/witness_smallest_L.json \
                                    ball_certificate_L1.8.txt
```

`make verify`, in the note directory, runs `note/verify_note_numbers.py`: it rebuilds every quantity printed above from the data files, scans this document for numeric literals, fails if any literal is not derived or is printed to more significant digits than its source carries, regenerates Appendix A's table from `certificates/witness.json` and compares it, and cross-checks the Gate A values against `REPORT.md`. Values attributed to another author are handled differently, and the difference is deliberate: they are *not* re-derived, and the verifier requires each to appear verbatim in `SOURCE_AUDIT.md` — the check is of provenance, that the number came from a reading of the source, not of correctness. The full suite is `./run_suite.sh`.

Licence

As the parent repository: code MIT, this note and the data CC BY 4.0. CC BY 4.0 requires attribution; if you use the note, the certificates or the data, please cite this note by its Zenodo DOI — version 1 is 10.5281/zenodo.23608288, which this version supersedes — and the parent paper [5] separately.

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